When it sends the `SIGTERM` signal to a Pod, Kubernetes will also inform its own components to stop forwarding requests to the terminating Pod. Since Kubernetes is a distributed system, and the two actions happen in parallel, there is a short time window when the terminating Pod might still receive requests, even if it has already started the graceful shutdown procedure. When that happens, those new requests will be rejected, resulting in errors in the clients. Our goal was to make the shutdown procedure transparent to the clients, so that scenario is unacceptable.

The recommended solution is to delay sending the `SIGTERM` signal to the Pod so that Kubernetes has enough time to spread the news across the cluster. By doing so, all Kubernetes components will already know not to send new requests to the Pod when it starts the graceful shutdown procedure. Technically, the delay can be configured through a `preStop` hook. Let's see how we can update the Deployment manifest for Catalog Service to support a transparent and graceful shutdown.

Open the deployment.yml file located in catalog-service/k8s, and add a `preStop` hook to delay the `SIGTERM` signal by 5 seconds.

Listing 7.4 Configuring a delay in Kubernetes before the shutdown starts

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: catalog-service
  labels:
    app: catalog-service
spec:
  ...
  template:
    metadata:
      labels:
        app: catalog-service
    spec:
      containers:
        - name: catalog-service
```